

MAWSON STATION, Antarctica

WMO: 895640

Lat: 67.602S

Lon: 62.875E

Elev: 52

StdP: 14.67

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-20.2	-17.4	-35.5	0.7	-18.5	-32.8	0.9	-16.2	83.8	11.2	78.6	11.7	13.7	160	1.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	38.5	32.3	36.9	31.1	35.7	30.4	32.9	37.4	31.8	36.0	30.9	35.1	18.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.6	22.6	33.6	27.0	21.0	33.1	25.6	19.7	32.7	12.2	37.5	11.8	36.2	11.4	35.2	37.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 73.2	65.2	57.9	-22.5	41.2	4.0	2.0	-25.4	42.7	-27.7	43.8	-29.9	45.0	-32.9	46.4		
(5)			-23.0	34.6	3.9	1.4	-25.8	35.6	-28.0	36.5	-30.2	37.3	-33.0	38.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	31.1	24.0	13.2	5.2	1.8	2.5	0.0	-0.8	1.6	8.6	22.4	30.3	
	DBStd	13.84	2.97	4.79	6.67	8.11	9.44	9.08	10.29	8.95	8.97	7.04	5.49	3.49	
	HDD50	14017	587	727	1140	1343	1495	1424	1550	1576	1453	1283	828	611	
	HDD65	19492	1052	1147	1605	1793	1960	1874	2015	2041	1903	1748	1278	1076	
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	27.4	22.1	27.5	29.4	28.0	28.5	29.4	29.5	28.9	29.1	26.7	26.7	22.8	
Precipitation	PrecAvg	12.2	1.3	1.2	1.0	0.9	0.9	1.0	1.1	0.9	0.8	0.9	1.1	1.1	
	PrecMax	21.3	3.1	2.4	1.7	2.2	2.1	3.3	2.7	4.3	1.9	2.6	2.4	2.3	
	PrecMin	6.2	0.2	0.3	0.4	0.2	0.2	0.2	0.3	0.2	0.0	0.1	0.1	0.2	
	PrecStd	3.4	0.7	0.5	0.4	0.5	0.5	0.6	0.7	0.7	0.5	0.6	0.7	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	38.5	31.2	26.0	28.4	24.2	24.5	22.6	23.9	27.8	35.5	41.0	
		MCWB	33.6	32.3	26.6	21.3	23.1	19.9	21.9	18.2	19.3	22.6	29.2	33.9	
	2%	DB	38.6	34.7	27.8	22.9	22.2	21.2	21.0	18.0	20.1	24.2	33.5	38.7	
		MCWB	32.4	29.5	23.7	19.4	18.3	17.6	18.3	15.2	16.7	19.9	28.3	32.4	
	5%	DB	37.2	33.0	25.3	20.0	18.5	17.9	17.7	15.4	17.4	21.5	32.0	37.0	
		MCWB	31.5	28.2	21.4	17.1	15.3	15.1	15.2	12.9	14.5	17.8	27.3	31.1	
	10%	DB	36.0	31.5	22.8	17.0	15.3	15.4	14.6	12.4	14.4	19.1	30.4	35.7	
		MCWB	30.7	26.9	19.3	14.3	12.9	12.7	12.2	10.0	11.8	15.7	26.1	30.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.7	32.8	27.4	22.3	23.2	20.5	22.6	18.7	19.7	23.2	30.6	34.6
			MCDB	39.5	36.4	30.3	25.4	28.2	23.8	23.9	21.3	23.1	27.1	33.6	39.9
2%		WB	33.1	30.5	23.9	19.7	18.6	17.7	18.7	15.3	17.0	20.2	29.0	33.0	
		MCDB	37.3	34.2	27.3	22.8	21.9	20.6	20.8	17.8	19.9	23.8	32.6	37.9	
5%		WB	32.1	28.6	21.7	17.1	15.5	15.3	15.2	12.8	14.5	18.0	27.8	31.7	
		MCDB	36.3	32.5	25.1	19.8	18.2	18.0	17.5	15.3	17.2	21.3	31.7	36.2	
10%		WB	31.2	27.0	19.5	14.3	12.9	12.8	12.2	10.0	11.8	15.7	26.3	30.7	
		MCDB	35.4	31.0	22.7	16.8	15.4	15.3	14.5	12.2	14.2	19.0	30.2	35.1	
Mean Daily Temperature Range		5% DB	MDBR	7.3	8.4	8.2	7.8	8.0	8.2	8.2	8.2	8.3	9.2	8.6	7.2
			MCDBR	7.8	8.1	8.2	8.2	8.7	9.3	7.8	8.6	8.3	9.8	8.5	8.0
	MCWBR		6.2	6.8	6.6	6.9	7.4	7.9	7.0	7.4	7.0	8.4	7.2	6.5	
	5% WB	MCDBR	7.0	7.5	7.4	7.8	8.4	9.0	7.7	8.4	8.1	9.5	8.0	7.2	
		MCWBR	5.8	6.5	6.3	6.6	7.3	7.8	7.0	7.4	7.0	8.2	7.0	6.0	
Clear-Sky Solar Irradiance	taub	0.257	0.239	0.212	0.166	0.117	N/A	0.119	0.162	0.198	0.210	0.239	0.258		
	taud	2.267	2.336	2.326	2.160	1.908	N/A	1.890	2.117	2.229	2.243	2.231	2.214		
	Ebn at Noon	317	306	284	227	87	0	72	224	283	315	322	321		
	Edh at Noon	36	29	22	12	3	0	3	13	23	31	37	40		
All-Sky Solar Radiation	RadAvg	2492	1725	895	310	39	0	11	167	674	1555	2419	2835		
	RadStd	123	125	67	16	2	0	1	8	28	54	66	102		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page